

Second Normal Form (2NF) — Relational Database Design

Definition: A relation is in **Second Normal Form (2NF)** if it is in **1NF** and **no non-prime attribute is partially dependent on any candidate key** of the relation. Every non-key attribute must be **fully functionally dependent** on the primary key.

1. Detailed Technical Explanation

Functional Dependency Concepts:

- **Full Functional Dependency:** $X \rightarrow Y$ is a full functional dependency if removal of any attribute A from X means the dependency no longer holds.
- **Partial Dependency:** Occurs when a non-prime attribute depends on only a *proper subset* of a composite primary key.

Conversion Example: 1NF to 2NF

Relation R(Student_ID, Course_ID, Student_Name, Course_Fee)

- **Candidate Key:** {Student_ID, Course_ID}
- **Functional Dependencies:**
 1. {Student_ID, Course_ID} $\rightarrow$ Student_Name, Course_Fee
 2. Student_ID $\rightarrow$ Student_Name (*Partial Dependency: Student_Name depends only on part of key*)
 3. Course_ID $\rightarrow$ Course_Fee (*Partial Dependency: Course_Fee depends only on part of key*)

Decomposition into 2NF:

We split table R into 3 normalized tables:

1. **STUDENT (Student_ID, Student_Name)**
 - Primary Key: **Student_ID**
2. **COURSE (Course_ID, Course_Fee)**
 - Primary Key: **Course_ID**
3. **ENROLLMENT (Student_ID, Course_ID)**
 - Primary Key: **{Student_ID, Course_ID}**
 - Foreign Keys: **Student_ID** refs STUDENT, **Course_ID** refs COURSE

2. Core Concepts & Memory Keywords

- **Non-prime Attribute:** An attribute that is not part of any candidate key.
- **Prime Attribute:** An attribute that belongs to at least one candidate key.
- **Partial Dependency Removal:** Splitting composite key attributes into separate lookup tables.

3. Must-Write Points for Exams

- 2NF is only relevant when the relation has a **composite candidate key** (candidate key containing 2 or more attributes).
 - Relations with a single-attribute primary key in 1NF are **automatically in 2NF**.
 - 2NF eliminates **update anomalies** caused by repeating non-key attribute values.
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4. Quick Recall Flow

1NF Relation -> Identify Partial Dependencies (Non-Prime -> Part of Key) -> Decompose Tables -> 2NF Achieved